# Center Estates ERP

CONSTRUCTION OVERSIGHT & FIELD TELEMETRY

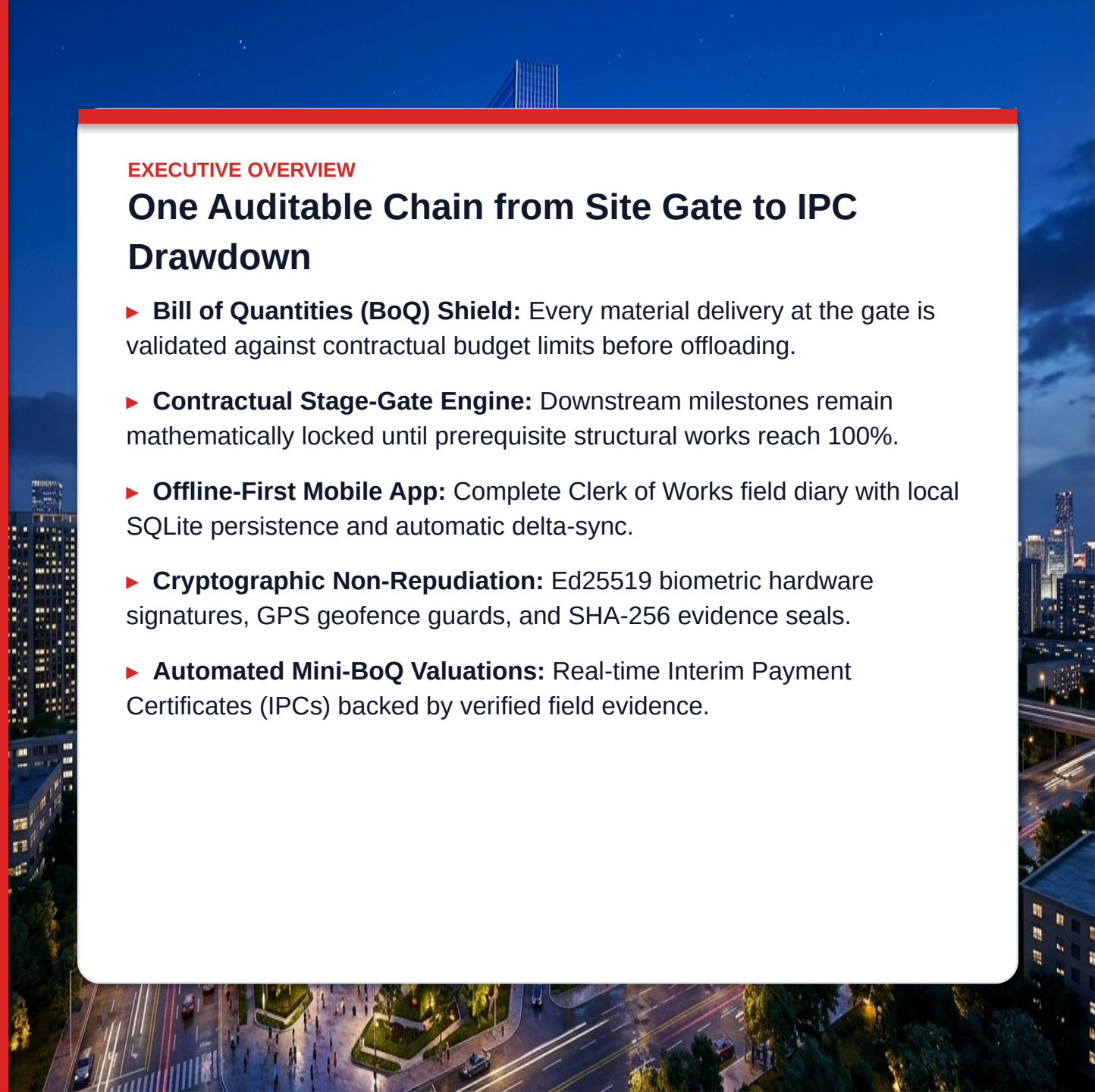
LIVE STAKEHOLDER BRIEFING

# Eliminating Capital Leakage in Real Estate Construction.

Connecting the physical site gate directly to bank drawdown approvals via BoQ Allowance Shields, Contractual Stage-Gates, and Offline-First Mobile Telemetry.

| 100% | SHA-256 | 0% Leakage |
|---|---|---|
| BoQ-Bound Gate Intake | Tamper-Evident Seals | Stage-Gate Protected |

Featured Project: Karam Apartments — Lang'ata (PRJ-KRM-01) • centerestates.com



EXECUTIVE OVERVIEW

## One Auditable Chain from Site Gate to IPC Drawdown

- **Bill of Quantities (BoQ) Shield:** Every material delivery at the gate is validated against contractual budget limits before offloading.
- **Contractual Stage-Gate Engine:** Downstream milestones remain mathematically locked until prerequisite structural works reach 100%.
- **Offline-First Mobile App:** Complete Clerk of Works field diary with local SQLite persistence and automatic delta-sync.
- **Cryptographic Non-Repudiation:** Ed25519 biometric hardware signatures, GPS geofence guards, and SHA-256 evidence seals.
- **Automated Mini-BoQ Valuations:** Real-time Interim Payment Certificates (IPCs) backed by verified field evidence.

01 • INDUSTRY CHALLENGE

# The Three Critical Blind Spots in Construction Execution

Why 15% to 25% of commercial development capital is lost between the site gate and the finance office.

## 01 Phantom & Over-Deliveries

Paper delivery notes signed at the site gate with zero live cross-reference to the approved Bill of Quantities.

- **Budget Overruns:** Materials delivered beyond contractual allowances without real-time visibility.
- **Supplier Opacity:** No standardized tracking of batch quality or approved manufacturers.
- **Retrospective Discovery:** Variances only uncovered months later during final account audits.

## 02 Premature Progress Claims

Contractors claiming completion on downstream finishes before critical structural foundations are certified.

- **Front-Loaded Claims:** Capital disbursed ahead of actual physical progress on site.
- **Subjective Valuations:** Visual estimates disconnected from measured BoQ sub-elements.
- **Dispute Exposure:** Lack of immutable stage-gate prerequisites triggers arbitration.

## 03 Disconnected Site Diaries

Clerk of Works observations scattered across paper logbooks, email threads, and unverified WhatsApp photos.

- **Unverified Media:** Site photos lack hardware GPS coordinates and tamper-evident hashes.
- **Missed Concrete Tests:** 7, 14, and 28-day cylinder crushing tests forgotten or unlinked.
- **Snag Amnesia:** Structural defects logged verbally fail to reach the QS before payment.

**GOVERNANCE GUARANTEE** **Traditional oversight relies on trust and post-mortem audits. Center Estates enforces proactive digital stage-gates.**

02 • UNIFIED PLATFORM ARCHITECTURE

# End-to-End Architecture: Mobile Field App ↔ REST API ↔ Web ERP

A single continuous operating record connecting field engineers, quantity surveyors, and project financiers.

## APP Mobile Field App (Expo / RN)

Authoritative source of physical site truth for Clerks of Works and Resident Engineers.

- **Offline-First SQLite:** TinyBase reactive store persists all logs & drafts without 4G.
- **Rugged Field UX:** Searchable BoQ picker and one-tap Kenyan supplier chips.
- **GPS Geofencing:** Hardware coordinate lock against project site boundaries.
- **Biometric Sign-Off:** Ed25519 hardware key signatures on every submission.

⇄

## API Central API (Django & MySQL)

Production governance engine running at api.centerestates.com.

- **Atomic Delta-Sync:** Idempotent /sync/pull and /sync/push endpoints.
- **Stage-Gate Validator:** Rejects progress >0% if prerequisite milestones <100%.
- **SHA-256 Evidence Seals:** Cryptographic hash computed over payload + photo bytes.
- **Role-Based Access:** Strict separation of Clerk, Consultant, and Owner permissions.

⇄

## WEB Office Web ERP (Next.js 14)

Executive command center for Developers, Project Managers, and QSs.

- **CPM Work Plan & Gantt:** Predecessor scheduling and critical path tracking.
- **Bill of Quantities (BoQ):** Live line-item budgets, rates, and variance shields.
- **Mini-BoQ Valuations:** Automated Interim Payment Certificates (IPCs).
- **Live Snag Kanban:** Instant visibility into field-reported quality defects.

**GOVERNANCE GUARANTEE** **Immutable Field Truth: Office users can review and certify, but can never alter or delete raw field evidence.**

03 • FIELD MATERIAL CONTROL

# Site Gate Deliveries Bound to BoQ Allowances

Preventing material over-delivery and budget leakage before the truck leaves the site perimeter.

### 01 Searchable BoQ Item Picker

Clerk selects directly from the project's approved Bill of Quantities (e.g. BOQ-02.04 Cement). Automatically binds item code, unit, and contractual rate.

### 02 Kenyan Supplier Quick-Chips

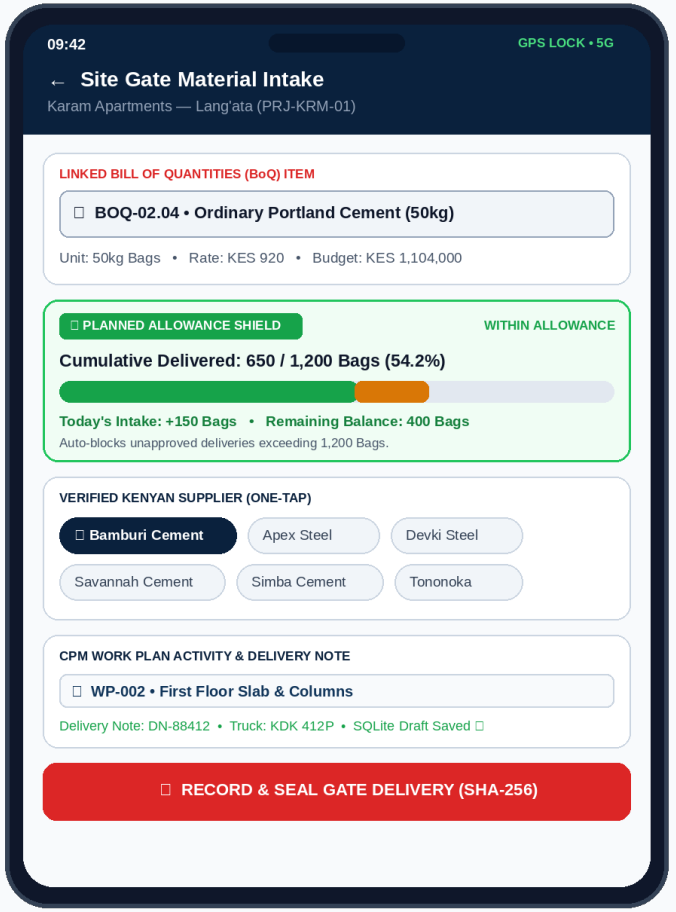
One-tap verified supplier chips (Bamburi, Apex Steel, Devki, Savannah, Simba, Tononoka) eliminate typos and standardize procurement audits.

### 03 Planned Allowance Shield

Live progress bar compares cumulative delivered quantity against total BoQ allowance. Triggers an immediate red warning if intake exceeds budget.

### 04 Work Plan Task Tagging

Every material intake is tagged to a specific CPM Work Plan activity (e.g. WP-002 First Floor Slab) and persists locally via useFormDraft.




**GOVERNANCE GUARANTEE** **100% Traceability: Every bag of cement and ton of steel is linked to a BoQ line item, supplier, and work plan task.**

04 • CONTRACTUAL GOVERNANCE

# The Stage-Gate Engine: Mathematical Payout Protection

Downstream milestones cannot advance above 0% until all prerequisite structural works reach 100%.

**MS-01: Substructure & Raft**
50% IN PROGRESS
Prerequisite Incomplete

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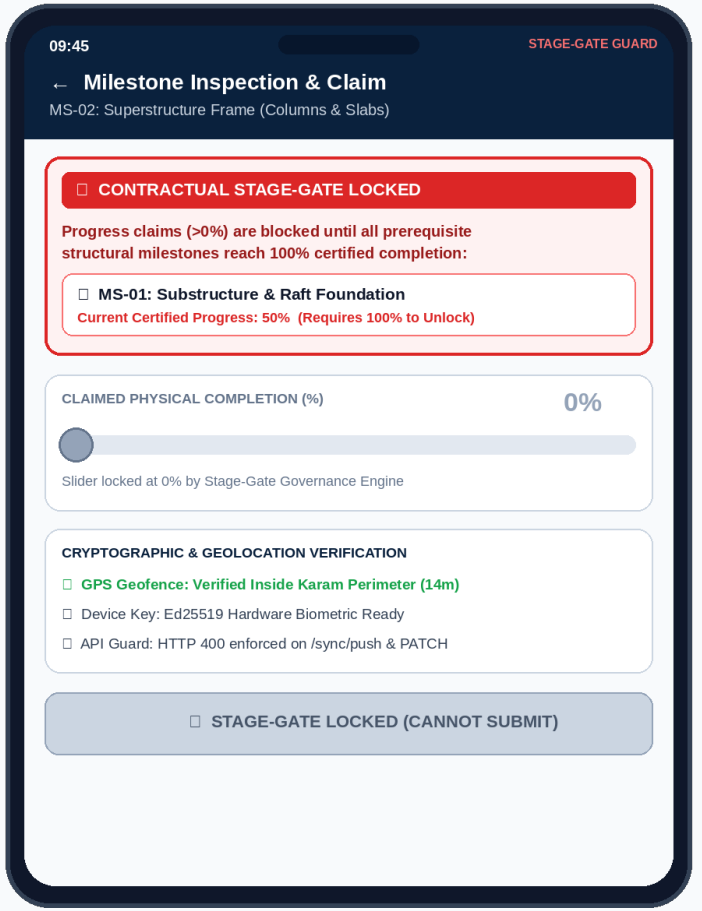
**MS-02: Superstructure Frame**
🔒 LOCKED AT 0%
Blocked by MS-01 (<100%)

→

**MS-03: Roofing & Finishes**
🔒 LOCKED AT 0%
Blocked by MS-02 (<100%)

## Dual-Layer Enforcement (Mobile UI + Backend API)

- **Mobile Stage-Gate Lock Banner:** MilestoneDetailScreen evaluates evaluateMilestoneGovernance() in real time, displaying the exact blocking predecessors and disabling the submit button.
- **Strict Backend Validation:** Both PATCH /milestones/<id>/ and POST /sync/push call milestone.get_incomplete_prerequisites() and return HTTP 400 if any predecessor is <100%.
- **Zero Premature Drawdowns:** Protects developers and lenders from paying for superstructure or finishes before foundational integrity is 100% certified.




**GOVERNANCE GUARANTEE** Stage-gates turn construction contracts into executable code—eliminating subjective certification bias.

05 • QUALITY, STRUCTURAL TESTING & HSE

# Field Snagging, Concrete Cube Lifecycle & Safety Escalation

Capturing structural quality tests and engineering defects at the source with live Kanban synchronization.

## Mobile Snag & Defect Ticketing

Resident Engineers log structural and finishing defects directly from the floor plate.

- **Severity & Category:** Urgent / High / Normal routing across Structural, MEP, and Finishes.
- **Geotagged Photo Proof:** Mandatory camera capture with hardware GPS coordinates.
- **Live Office Kanban:** Syncs immediately to the Web ERP oversight board for contractor resolution.
- **Closure Verification:** Requires before-and-after photographic sign-off.

## Concrete Cube Curing Scheduler

Automated structural pour compliance aligned with BS 8110 / Eurocode 2 standards.

- **Batch & Slump Intake:** Logs concrete grade (e.g. C30/37), mixer truck, and slump test results.
- **Automated 7 / 14 / 28-Day Alerts:** Automatically schedules mandatory lab crushing tests.
- **Pass/Fail Thresholds:** Highlights compressive strength shortfalls (MPa) before slab loading.
- **Permanent Audit Archive:** Lab certificates attached directly to the structural milestone.

## HSE & Regulatory Compliance

Immediate escalation protocol for site safety inspections, near-misses, and NCA/DOSHS compliance.

- **PPE & Scaffold Audits:** Daily digital checklists completed by the Clerk of Works.
- **Incident Root-Cause Tracking:** Severity grading and Corrective Action Plan (CAP) workflows.
- **Statutory Readiness:** Maintains full National Construction Authority (NCA) inspection logs.
- **Zero-Delay Escalation:** High-severity incidents trigger instant executive alerts.

**GOVERNANCE GUARANTEE** **Quality assurance records and concrete test certificates form an immutable structural passport for handover.**

06 • COMMERCIAL & FINANCIAL CONTROL

# Earned Value S-Curve & Automated Mini-BoQ Valuations (IPCs)

Converting verified field progress into audit-ready Interim Payment Certificates in minutes.

QS

## 1. Mini-BoQ Line-Item Valuation

Quantity Surveyors inspect the exact BoQ sub-elements linked to each milestone, calculating certified KES earned value with zero spreadsheet formulas.

AI

## 2. AI Progress & Variance Audit

Automated variance engine compares Planned Value (PV), Earned Value (EV), and Actual Cost (AC), flagging schedule slippage and material anomalies.

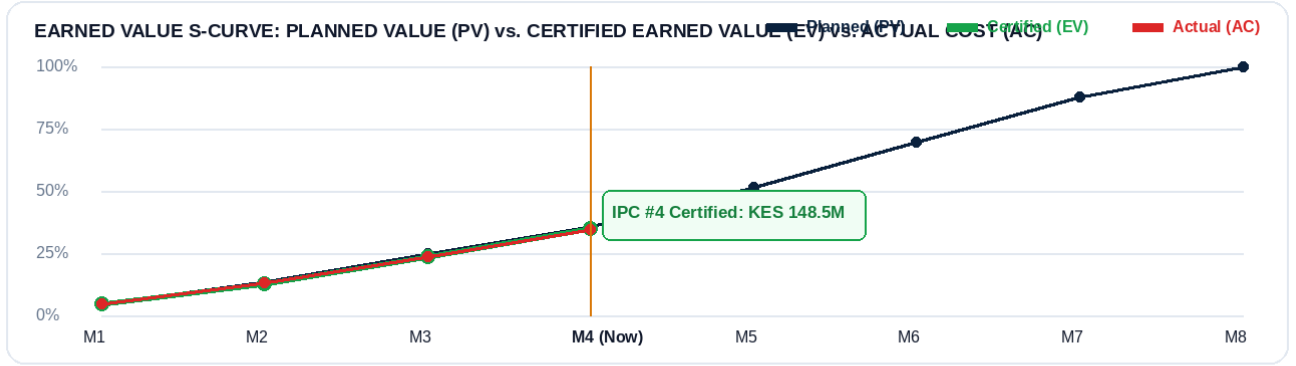
IPC

## 3. Escrow & Retention Drawdowns

Generates bank-ready PDF Interim Payment Certificates with automatic 10% retention deductions and cryptographic links to supporting site photos.

Center Estates Web ERP • Karam Apartments — Lang'ata (PRJ-KRM-01) • Commercial & IPC Valuation

| CONTRACT SUM (BoQ) | CERTIFIED TO DATE (IPC #4) | RETENTION HELD (10%) | MATERIAL VARIANCE |
|---|---|---|---|
| KES 420.0M | KES 148.5M | KES 14.85M | -0.4% (SAVING) |
| Approved Baseline | 35.4% Earned Value | Escrow Protected | BoQ Shield Active |




| BoQ CODE | ELEMENT DESCRIPTION | BUDGET (KES) | VERIFIED % | CERTIFIED IPC | AUDIT STATUS |
|---|---|---|---|---|---|
| BOQ-01.01 | Substructure & Raft Foundation | 84,000,000 | 100.0% | 84,000,000 | SHA-256 Sealed |
| BOQ-02.04 | Superstructure RC Frame (Fl 1-3) | 112,000,000 | 57.6% | 64,500,000 | GPS + Photo Verified |
| BOQ-03.01 | Roofing & Structural Waterproofing | 48,000,000 | 0.0% | 0 | Stage-Gate Locked |

**GOVERNANCE GUARANTEE** Every shilling certified on an Interim Payment Certificate is backed by geotagged photos and sealed gate deliveries.

07 • ANTI-FRAUD ARCHITECTURE

# Offline Resilience & Cryptographic Non-Repudiation

Purpose-built for deep basements and remote sites without sacrificing bank-grade security.

## 01 Offline SQLite Engine

Deep excavations and basements rarely have reliable 4G.

- **Local Persistence:** All BoQ items, milestones, and daily logs cached in SQLite.
- **Draft Recovery:** useFormDraft hook prevents data loss if battery dies mid-inspection.
- **Auto Delta-Sync:** Queued payloads sync atomically once connectivity returns.

## 02 Biometric Hardware Keys

Eliminates shared passwords and unauthorized proxy sign-offs.

- **Ed25519 Keypair:** Generated inside the mobile device's secure hardware keystore.
- **Biometric Prompt:** Fingerprint / FaceID required to sign milestone submissions.
- **Non-Repudiation:** Cryptographic proof of the exact engineer who certified the work.

## 03 GPS Geofence Guard

Prevents off-site photo uploads and Recycled Photo fraud.

- **Perimeter Lock:** Compares hardware GPS coordinates against project polygon.
- **Distance Telemetry:** Records exact meter offset and GPS accuracy radius.
- **Fraud Flagging:** Out-of-bounds submissions are automatically flagged for audit.

## 04 SHA-256 Evidence Seals

Tamper-evident cryptographic sealing at the moment of capture.

- **Composite Hash:** Computed over milestone ID, progress %, notes, GPS, and photo bytes.
- **Server Verification:** Re-verified upon ingestion by api.centerestates.com.
- **Audit Certificate:** Displayed on Web ERP alongside the green Verified badge.

**GOVERNANCE GUARANTEE** **Security is embedded directly into the mobile data capture layer—making fraudulent progress claims mathematically impossible.**

08 • LIVE DEMONSTRATION SCRIPT

# Recommended 5-Step Live Demonstration Flow (8 Minutes)

A high-impact, cohesive walkthrough using Karam Apartments — Lang'ata (PRJ-KRM-01).

| | | | |
|---|---|---|---|
| STEP 01 | **Office Baseline Setup (Web ERP)** | Open Karam Apartments (PRJ-KRM-01). Show the CPM Work Plan Gantt and approved BoQ allowances. | 1.5 MIN |
| STEP 02 | **Site Gate Delivery (Mobile App)** | Open Daily Log -> Material Delivery. Pick 'BOQ-02.04 Cement', tap 'Bamburi', show Planned Allowance Shield. | 2.0 MIN |
| STEP 03 | **Stage-Gate Lockout (Mobile App)** | Open 'Superstructure Frame'. Highlight the red '🔒 Contractual Stage-Gate Locked' banner blocking claims. | 1.5 MIN |
| STEP 04 | **Live Defect Sync (Mobile -> Web)** | Capture a rebar spacing snag on Mobile. Switch to Web ERP to show immediate appearance on the Kanban board. | 1.5 MIN |
| STEP 05 | **Mini-BoQ & IPC Valuation (Web)** | Open Milestones -> Mini-BoQ Valuation. Inspect certified KES value, retention deduction, and AI Progress Audit. | 1.5 MIN |

GOVERNANCE GUARANTEE **Keep the entire demonstration anchored inside Karam Apartments (PRJ-KRM-01) to prove end-to-end data continuity.**

09 • BUSINESS CASE & ROI

# Measurable Financial Impact Across Every Project Stakeholder

Why institutional developers, consultants, and project financiers standardize on Center Estates.

## DEV Real Estate Developers

Protects equity capital and accelerates project delivery.

- **15%–20% Cost Protection:** Eliminates phantom deliveries and unapproved BoQ overages.
- **Real-Time Visibility:** Live executive S-Curve across all active development sites.
- **Seamless Handover:** Completed units transition directly into Center Estates Sales & Lettings.

## QS Project Managers & QSs

Replaces manual spreadsheet reconciliation with automated governance.

- **80% Faster IPC Valuations:** Mini-BoQ calculations generated in minutes instead of days.
- **Standardized Site Records:** Automated concrete cube schedules and daily weather/labour logs.
- **Dispute Defense:** Time-stamped, geotagged photographic proof resolves contractor claims.

## FIN Banks & Project Financiers

De-risks construction debt drawdowns with cryptographic proof.

- **Verified Drawdowns:** Every tranche disbursement tied to 100% stage-gated milestones.
- **Anti-Collusion Controls:** Ed25519 biometric signatures + GPS geofence validation.
- **ESG & Statutory Audit:** Complete NCA, DOSHS, and structural quality compliance trail.

**GOVERNANCE GUARANTEE** **Center Estates transforms construction monitoring from a passive reporting chore into an active financial shield.**

10 • SUMMARY & LIVE Q&A

# One Operating Record: From Groundbreaking to Estate Handover

The mobile app is live, the backend is synchronized, and Karam Apartments is ready for inspection.

GATE INTAKE (BoQ) → FIELD VERIFICATION → STAGE-GATE GUARD → IPC CERTIFICATION → ESTATE HANDOVER

## Ready for Live Walkthrough & Stakeholder Scenarios

### Live Demonstration Environment

- **Web ERP Command Center:** https://centerestates.com
- **Presentation & Resource Hub:** https://centerestates.com/presentations/
- **Featured Project:** Karam Apartments — Lang'ata (PRJ-KRM-01)
- **Automated Test Suite:** 100% Pass Rate (135 Web + 73 Mobile Tests)

### Platform & Engineering Governance

- **Suite:** Center Estates Enterprise Construction & Field Mobility
- **Architecture:** Next.js 14 Web • Expo React Native • Django REST API
- **Security:** Ed25519 Biometrics • SHA-256 Seals • SQLite Offline Sync
- **Organization:** Center Estates Enterprise Engineering

GOVERNANCE GUARANTEE

**Thank you. Let us open Karam Apartments (PRJ-KRM-01) for the live demonstration.**